

Improving Forest Carbon Models under Climate Change Using Drones and Experiments

Context: Anthropogenic warming is reshaping natural systems globally^{1,2}, and its most frequently observed biological impact is a shift in the timing of recurring life history events (phenology)^{3,4,5,6,7}. Spring phenology (e.g. budburst and leafout) has advanced by 0.5⁸ to 4.2 days/decade^{9,10}, driven mainly by temperature^{11,4,12}; autumn events (e.g. budset and leaf colouring) are delayed but to a much lesser extent^{13,14} and driven by shortening photoperiod^{15,16,17} and cooling temperatures^{15,18}. Together these shifts lengthen the growing season, and the long-standing assumption is that tree growth should increase accordingly. Yet a growing body of work has failed to connect growth and growing season length^{19,20,21,22,23}: Dow *et al.* (2022) showed that despite an earlier growth onset, neither growth rate nor annual increment rose with season length. Because temperate forests account for 40% of the global net carbon sink²⁴, this paradox propagates directly into carbon-cycle projections and their feedbacks to future climate^{25,26}.

Resolving it requires asking why trees do not grow more when the season is longer. I hypothesize two non-exclusive classes of limit: external (environmental)²⁷ and internal (physiological)²⁸. External drivers—drought, spring frost and heat waves—are difficult to isolate in natural settings because they co-occur and act on whole communities rather than on individuals^{29,30,31,32}; experiments are therefore essential to tease apart, for example, a warm spring from a severe drought later in the same season^{33,34}, and to refine forest carbon sequestration projections^{35,23}. Internal limits arise instead from how trees allocate carbon between growth and storage: if growth is sink- rather than source-limited, a longer season yields more photosynthate but no more wood; the surplus accumulates as nonstructural carbohydrate reserves^{28,36}, and any benefit may surface only in a later year, if at all³⁷. Testing this demands multi-year experimental data, which do not yet exist. Experiments are also usually run on juvenile trees, whose responses inform regeneration projections but translate poorly to the mature trees holding the overwhelming majority of forest biomass^{38,20,39}. Closing that gap in situ is now tractable: UAV multispectral imagery paired with machine learning yields sample sizes far beyond traditional ground observation, at finer spatial and temporal resolution than satellites^{40,41,42}. I therefore propose two experiments testing internal (Chapter 1) and external (Chapter 2) limits to growth, alongside a large-scale observational project on mature mixed-forest canopies (Chapter 3).

Chapter 1. Lag effects of extended growing seasons—internal limits: Shifts in phenology have consequences for growth within the current season—a hypothesis I tested successfully in my MSc thesis—but no experiment has yet quantified whether longer seasons carry lag effects into subsequent years^{36,37,43}. *Objective:* test whether the growth response to season length is contingent on the previous year's conditions. *Hypothesis:* trees that extend activity in one year draw down carbon reserves, offsetting growth gains in the following year, with the offset strongest in species that invest heavily in storage. In fall 2025 I completed a large-scale growing season length experiment using 15 replicates of seven species over two consecutive years, assessing species' potential to stretch their activity schedules and whether this translates into increased growth across seasons. In year 1 of the award, I will analyse the tens of thousands of observations collected in 2024 and 2025 using a joint Bayesian hierarchical model estimating phenology and growth simultaneously. The model will partially pool across species and propagate uncertainty in estimated phenological dates directly into the growth component. Current studies treat those dates as fixed covariates, a simplification that biases effect sizes. I will develop it through an ongoing collaboration with Dr. Charles Margossian, a statistician and expert in Bayesian inference, building on the Stan workflow I established during my MSc. Though mostly computational, this chapter will deliver the modelling tool called for by Wolkovich *et al.* (2025) to test multi-year growth dependence in natural systems.

Chapter 2. Drought and spring frost experiment—external limits: Climate change will shift not only growing season length but the timing of moisture deficits and the frequency of late-spring frosts⁴⁴.

Tree-ring research shows both can cause important tissue loss^{45,46,47}, yet whether affected trees also grow less—and whether the timing of the stress matters more than its intensity—remains unresolved^{48,46,49,50}. *Objective*: quantify how the timing of drought and of spring frost alters seasonal growth across species with contrasting life history strategies. *Hypothesis*: growth loss scales with the proximity of the stress to a species' peak growth period, so that an early-season drought is largely compensated while a solstice drought is not. Starting in year 2 of the award, I will use 15 replicates of 16 deciduous North American species (8 congeneric pairs, to avoid confounding effects of shared evolutionary history) spanning the full acquisitive–conservative gradient, across five treatments and a control, for a total of 1440 individuals. For the drought treatments, trees will be moved to growth chambers warmer and drier than ambient conditions to maximize evapotranspiration until soil moisture probes installed in every pot indicate each species' wilting point (the value at which soil water is no longer extractable by the plant), then returned to ambient conditions under constant irrigation. The three droughts differ only in timing: just after leafout; one week before the solstice, the period of peak growth for many species^{51,52,53}; and near the end of the season, just before growth cessation. For the frost treatments, trees will be forced to budburst in warm chambers and then held at -4°C for one hour—the first treatment at budburst, the second once leaves are fully elongated—following successful methods from Chamberlain *et al.* (2021). Across all treatments I will monitor phenology twice a week throughout the season, estimate biomass at the start and end of the season using allometric equations validated on a destructively harvested subset, and equip a subset of trees with magnetic dendrometers to resolve the exact timing of growth change in response to treatments. Analyses will use the hierarchical framework built in Chapter 1, letting me test whether timing effects are consistent across life history strategies or species-specific.

Chapter 3. Growth timing × phenology from UAV imagery: How closely growth tracks leaf phenology, and how that synchrony differs among species, is critical to forest carbon accounting yet has received little attention^{54,55}. *Objective*: quantify the relationship between leaf phenology and growth seasonality for individual canopy trees in a mixed forest. *Hypothesis*: species differ systematically in this coupling along a life history gradient, making canopy phenology a biased proxy for growth in carbon-cycle models. Across the three years of the award, I will fly UAVs equipped with multispectral sensors over a mixed forest at the St-Hippolyte biological research station (Qc), using cutting-edge machine learning methods^{56,57}. This site lets me follow up on work previously done by my laboratory¹⁶ and build a partnership with Dr. Étienne Laliberté, whose laboratory currently works there⁵⁸. I will fly every three to five days through the spring and autumn transitions and weekly at midseason, and use BalSAM, a model that segments tree crowns accurately and efficiently from repeated UAV images⁴², to track individual crowns across flights and infer start and end of season for each species and individual^{40,59}. I will pair this with 50 dendrometers distributed across the site recording stem diameter change on canopy trees. Combining both at high spatial and temporal resolution will let me robustly test the phenology–growth relationship at the individual level, improving both the data carbon-cycle models depend on^{25,26} and the remote sensing methods used to produce them.

Outreach and EDI: Extended seasons may benefit some species and harm others, reshaping forest dynamics across North America with feedbacks to future climate. I will continue contributing to the Tree Spotters program, which I have worked with since 2024, and will start a new phenology monitoring community science program in Montreal, recruiting in boroughs such as Montréal-Nord and Saint-Michel where tree canopy cover, and access to research participation, are lowest. For Chapter 3, I will approach the Mohawk Council of Kanehsatà:ke, whose members hunt and fish at Tioweró:ton (Doncaster 17) some 35 km north of the station, to discuss how seasonal indicators held in Indigenous knowledge might be recorded alongside the phenological metrics described above. Code and data will be archived publicly on the Knowledge Network for Biocomplexity, and results returned to participants in French and English.